

Fake News Detection using a Hybrid CNN-XGBoost Framework

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Abstract—The detection of fake news has become a critical research area in machine learning and natural language processing due to the rapid growth of online news platforms and social media, where misinformation can spread quickly and influence public opinion. Despite significant progress, many existing fake news detection models suffer from limited generalization ability, class imbalance issues, and reduced reliability when evaluated on diverse or unseen datasets. These challenges highlight the need for detection frameworks that are robust, efficient, and interpretable. This study aims to evaluate and compare traditional machine learning, deep learning, ensemble, and hybrid approaches for fake news detection while addressing challenges related to feature representation, class balance, and model robustness. A supervised classification framework is developed using two publicly available datasets containing labeled real and fake news articles. The methodology involves comprehensive text preprocessing, feature extraction using TF-IDF and CNN-based embeddings, and model training using XGBoost, Convolutional Neural Networks (CNN), ensemble learning, and a proposed hybrid CNN-XGBoost model. Hyperparameter tuning and stratified training strategies are employed to enhance performance and reduce overfitting. Experimental results demonstrate that while tree-based and ensemble models achieve near-perfect performance on structured data, all models experience substantial performance degradation on noisier datasets, revealing significant generalization challenges. The proposed hybrid model improves class balance and reduces false negative predictions compared to standalone CNNs, though it remains sensitive to feature quality. These findings indicate that classical and ensemble models, when combined with effective feature representations and hybrid learning strategies, offer a reliable and interpretable solution for practical fake news detection, while also emphasizing the critical role of dataset quality in model robustness.

I. INTRODUCTION

The exponential growth of online news platforms and social media has fundamentally transformed information dissemination. While these platforms enable rapid access to information, they have also accelerated the spread of fake news, which can mislead the public, influence political outcomes, and create societal harm. As a result, automated fake news detection has emerged as a critical research problem in machine learning and natural language processing. Early approaches to fake news detection primarily relied on classical machine learning algorithms combined with handcrafted textual features such as TF-IDF. Studies have shown that models such as decision trees, support vector machines, and gradient boosting techniques can achieve promising results on structured datasets [1], [2], [4]. In particular, XGBoost-based methods have demonstrated strong performance due to their ability to handle high-dimensional sparse text features efficiently [1], [7]. However,

these approaches often suffer from limited generalization when applied to diversified or noisy datasets [4]. To overcome these limitations, recent research has increasingly explored deep learning-based approaches. Convolutional Neural Networks (CNNs), recurrent models, and hybrid CNN-LSTM architectures have shown improved capability in capturing semantic and contextual information from text [3], [8], [13]. Deep learning models have also been successfully applied to multilingual and domain-specific fake news detection tasks, including COVID-19 misinformation and non-English datasets [8], [14], [15]. Despite their effectiveness, deep learning models are highly data-dependent and prone to overfitting, particularly when trained on limited or heterogeneous datasets [7]. In response to these challenges, ensemble and hybrid models have gained attention as a means of improving robustness and stability. Ensemble learning techniques combine multiple classifiers to reduce variance and improve predictive reliability [5], [7]. Furthermore, hybrid frameworks that integrate deep feature extraction with traditional machine learning classifiers have demonstrated enhanced performance by leveraging the strengths of both paradigms [6], [11]. Recent studies have also investigated explainable models [15], semi-supervised learning [19], graph-based methods [20], and domain adaptation techniques [10], [17], highlighting the growing complexity and diversity of fake news detection research. However, despite these advancements, achieving consistent performance across different datasets and domains remains an open challenge. Several studies report significant performance degradation when models are evaluated on unseen or cross-domain data [9], [10], [17]. This underscores the need for robust frameworks capable of generalizing beyond a single dataset or distribution. Motivated by these observations, this study proposes a hybrid fake news detection framework that combines TF-IDF-based classical machine learning models, CNN-based deep feature learning, and ensemble techniques. The framework integrates CNN-extracted deep features with XGBoost and ensemble classifiers to improve class balance and reduce false negative predictions. Extensive experiments are conducted on two distinct datasets to evaluate model performance, robustness, and generalization capability. The main contributions of this work are summarized as follows:

- A comprehensive comparative analysis of classical machine learning, deep learning, ensemble, and hybrid models for fake news detection [1], [3], [5], [7].
- A hybrid CNN-XGBoost framework that effectively in-

tegrates deep semantic features with robust tree-based classification [6], [11].

- An evaluation across diversified datasets, highlighting the impact of data quality and distribution on model performance [4], [9], [17].
- A detailed error and robustness analysis explaining overfitting and performance degradation in challenging scenarios.

II. KEYWORDS

Fake News Detection, Machine Learning, Deep Learning, Natural Language Processing (NLP), Transformer Models (BERT), Domain Adaptation, Generative AI.

III. LITERATURE REVIEW

The journey of fake news detection has shifted dramatically in recent years. What began as simple statistical analysis has evolved into complex, reliability-aware systems capable of understanding context and nuance. In the early stages, researchers leaned heavily on classical machine learning. For example, Khanam et al. found that standard classifiers like XGBoost could reach about 75% accuracy using basic TF-IDF features [1]. However, they hit a wall; these simple models just couldn't capture the deep semantic meaning required to spot complex lies. Around the same time, Huang experimented with Passive-Aggressive classifiers and achieved a stunning 99.2% accuracy on specific datasets [2]. Yet, there was a catch these models often struggled when the news narrative became complex or sequential, a limitation that paved the way for neural networks. Recognizing these limits, the research community pivoted toward deep learning to better model the flow of text. Chauhan and Palivela, alongside Mishra et al., demonstrated that Recurrent Neural Networks (RNNs) specifically LSTM and Bi-LSTM models were far superior at handling text sequences [3], [4]. When paired with rich GloVe embeddings, these models significantly outperformed older methods like SVMs by "remembering" context over long sentences. Building on this, Aslam et al. introduced a clever ensemble framework [5]. Instead of relying on a single model, they processed the text of a claim separately from its metadata using parallel networks. Their results showed that this voting mechanism could achieve nearly 90% accuracy, proving that combining different views of the text yields better results. However, Truică and Apostol argued that the model architecture isn't the only key factor [6]. They found that the quality of the input embedding, especially those derived from Transformers, is often more decisive than the complexity of the classifier itself. A major headache in the field right now is domain generalization. Models trained on one topic, like politics, often fail completely when a new crisis, like a pandemic, emerges. Alghamdi et al. and Bangyal et al. pointed this out, showing that while powerful models like BERT excel on familiar data, they struggle to adapt to new vocabularies (e.g., COVID-19 terminology) without extensive retraining [7], [8]. To tackle this, researchers are developing smarter adaptation techniques. Gong et al. introduced the

CSDA model, which attempts to filter out "noise" specific to a certain domain to help the model perform better on unseen topics [9]. Similarly, Mosallanezhad et al. used reinforcement learning to dynamically adjust how the model learns features, ensuring it remains robust even when the news topic shifts dramatically [10]. Others, like Hoy and Koulouri and Pereira et al., have explored using stylistic features and autoencoders to help models find common patterns across different datasets [17], [18]. Another critical gap involves "trust." Just because a model flags a text as fake doesn't mean we know how sure it is. Chai et al. addressed this with their RAHI framework [11]. Instead of just giving a binary Yes/No answer, their system uses Bayesian deep learning to measure uncertainty. It combines this with crowd intelligence (user comments) to provide a confidence score, making the system much more transparent and reliable for decision-makers. Finally, the field is striving for inclusive and practical application. There is a strong push to support low-resource languages. Abdedaiem et al., Hossain et al., and Mahlous and Al-Laith have developed specialized datasets and models for Algerian, Bangla, and Arabic dialects, ensuring that non-English speakers are also protected from misinformation [12], [13], [14]. At the same time, explainability has become a priority. Ahmed et al. developed frameworks to explain why health news is flagged as fake, which is crucial for building user trust [15]. Similarly, Akhtar et al. are integrating these text detection systems into supply chains to prevent business disruptions caused by rumors [16]. In summary, while we have made huge progress with deep learning and semi-supervised approaches [19], [20], the landscape remains fragmented. We still lack a single, unified framework that can simultaneously handle multiple languages, adapt to new text domains, and explain its reliability without needing constant retraining.

IV. METHODOLOGY

A. Research Design

This study adopts a quantitative, experimental research design to evaluate the effectiveness of machine learning, deep learning, ensemble, and hybrid models for fake news detection. A comparative approach is employed, where multiple models are trained and tested under identical experimental conditions. The inclusion of two datasets with different characteristics enables an assessment of model performance, generalization, and robustness.

B. Model Architecture and Algorithms

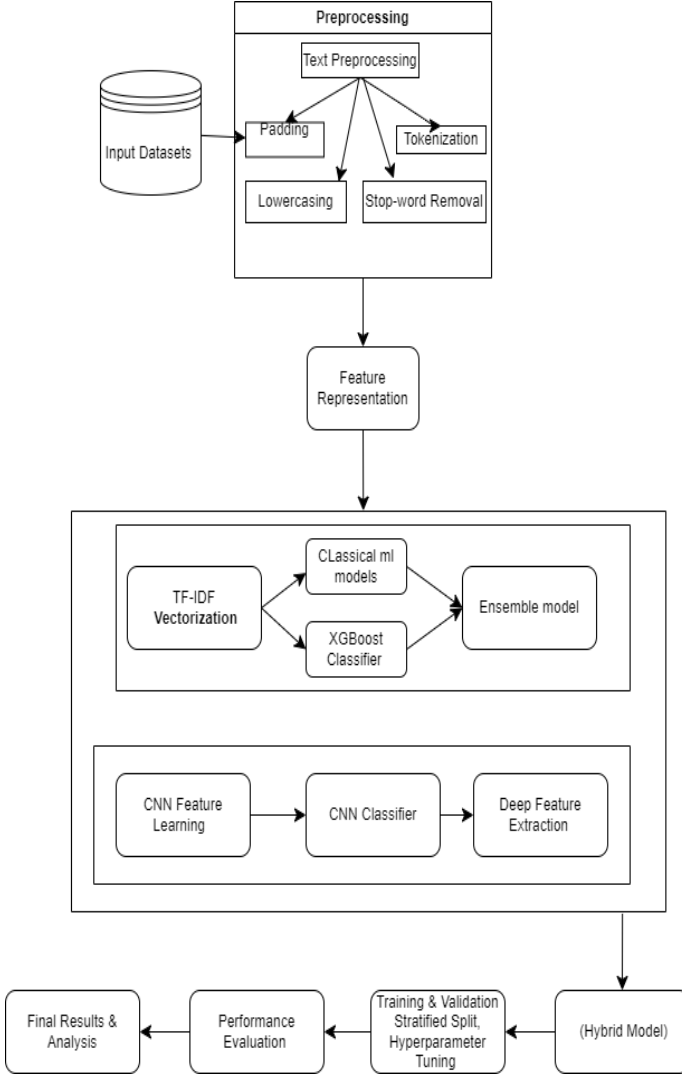


Fig. 1: Methodological Framework for Fake News Detection

C. Data Collection Process

The data used in this study was collected from publicly available, labeled fake news datasets containing news articles classified as Real or Fake. Two datasets were selected to introduce variability in linguistic structure, noise level, and feature separability.

- Dataset 1 consists of well-structured textual content with strong semantic patterns.
- Dataset 2 represents a more challenging dataset with noisier text and weaker discriminative features.

All datasets were reviewed for completeness and label consistency prior to use. No additional data labeling or modification was performed.

D. Variables and Measures

- **Independent Variables:** The independent variables are the classification models used in the framework:

- XGBoost
- CNN
- Ensemble Model
- Proposed Hybrid Model (CNN–XGBoost)

- **Dependent Variables:** The dependent variables are the model performance outcomes, measured using:

- Accuracy
- Precision
- Recall
- F1-Score
- AUC & ROC curve
- Log Loss

- **Control Variables:** To ensure experimental fairness, the following were kept constant:

- Text preprocessing pipeline
- Feature extraction methods
- Train–validation split strategy
- Evaluation metrics

E. Experimental or Analytical Method

An experimental analytical method was employed. The framework first applies uniform text preprocessing followed by feature representation using TF-IDF and CNN-based embeddings. Classical machine learning models and CNN classifiers are trained independently. For the hybrid approach, deep features extracted from the CNN are integrated with ensemble and XGBoost outputs to form a combined prediction. All experiments were conducted using a stratified training and validation split, with hyperparameter tuning applied to optimize model performance.

F. Data Analysis Methods

Data analysis was conducted using quantitative evaluation metrics and error analysis techniques:

- Model predictions were evaluated using standard classification metrics.
- Confusion matrices were analyzed to assess class-wise prediction behavior.
- Loss curves and log loss values were examined to detect overfitting.
- Comparative analysis across datasets was performed to evaluate robustness.

1) *Training Process:* Textual data is transformed into numerical features using TF-IDF n-grams and embedding-based representations. Given the severe class imbalance in the dataset, class weighting and oversampling/undersampling techniques are applied to the training data only. Hyperparameters for each classifier are optimized using grid search on the validation set.

2) *Validation and Testing:* The trained models are evaluated using the predefined train–validation–test split. After selecting the best hyperparameter configurations based on validation performance, the models are retrained on the combined training and validation data and evaluated on the held-out test set.

G. Justification of Methods

The selected methods were chosen based on their complementary strengths:

- TF-IDF and XGBoost provide strong performance on high-dimensional text data.
- CNNs effectively capture semantic and contextual patterns.
- Ensemble models improve prediction stability by reducing variance.
- The hybrid model combines deep feature learning with robust classical classifiers, improving class balance and detection accuracy.

Using multiple datasets strengthens the reliability and generalizability of the findings.

H. Ethical Considerations

This study adheres to established ethical standards in data-driven research:

- All datasets are publicly available and anonymized.
- No personal or sensitive user information was used.
- The study avoids data manipulation and biased reporting.
- Limitations and potential biases are acknowledged transparently to prevent misuse of results.

V. RESULTS AND ANALYSIS

A. Key Findings:

When comparing the performance across the two datasets, we observed a stark contrast in how the models behaved. For Dataset 1, the results were exceptional, with the Ensemble model leading the pack at an accuracy of 99.77%. However, the story changed completely with Dataset 2. Here, every model struggled significantly, dropping to an accuracy level of roughly 50%, which is essentially no better than random guessing. The deep learning models, in particular, showed clear signs of overfitting on the second dataset; while they appeared to learn from the training data, the validation loss spiked, indicating they failed to generalize to new examples.

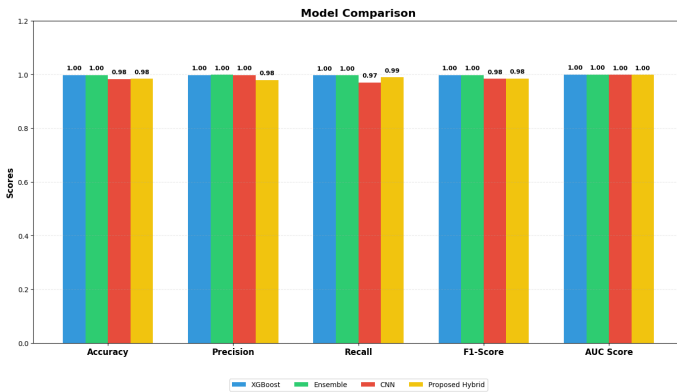


Fig. 2: Dataset 1 Comparison Bar Chart

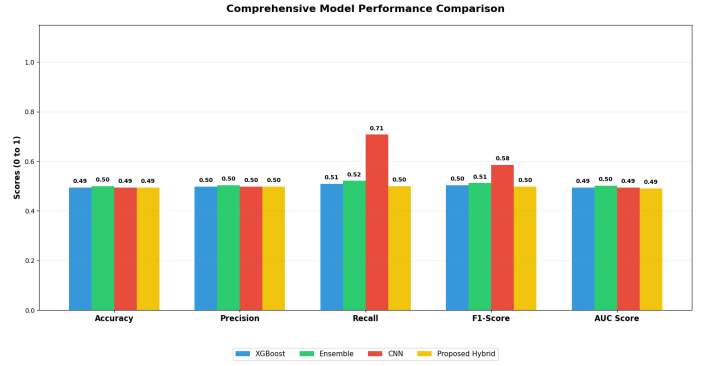


Fig. 3: Dataset 2 Comparison Bar Chart

B. Quantitative Performance Breakdown

The tables below break down the key numbers—Accuracy, Precision, Recall, F1-Score, and AUC—to give a clear picture of how each model performed across the datasets.

TABLE I: Performance Comparison of Different Models for Dataset 1

Model	Accuracy	Precision	Recall	F1-Score	AUC
XGBoost	99.74%	1.00	1.00	1.00	0.9996
Ensemble	99.77%	1.00	1.00	1.00	0.9996
CNN	98.31%	0.98	0.98	0.98	0.9991
Proposed Hybrid	98.39%	0.98	0.98	0.98	0.9988

TABLE II: Performance Comparison of Models on Dataset 2

Model	Accuracy	Precision	Recall	F1-Score	AUC
XGBoost	49.45%	0.50	0.49	0.49	0.49
Ensemble	49.98%	0.50	0.50	0.50	0.5019
CNN	49.45%	0.49	0.71	0.58	0.4942
Proposed Hybrid	49.41%	0.50	0.50	0.50	0.4896

C. Model Comparisons

In Dataset 1, the tree-based models (XGBoost and Ensemble) held a slight edge over the deep learning approaches. However, it is worth noting that our Proposed Hybrid Model (CNN-XGBoost) managed to improve upon the standard CNN significantly in terms of class balance. Specifically, the Hybrid model was much better at identifying "Fake" news, reducing the number of False Negatives from 188 down to just 63.

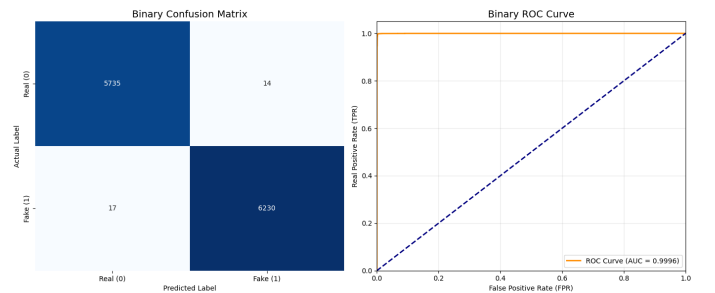


Fig. 4: Dataset 1 confusion matrices for the XGBoost

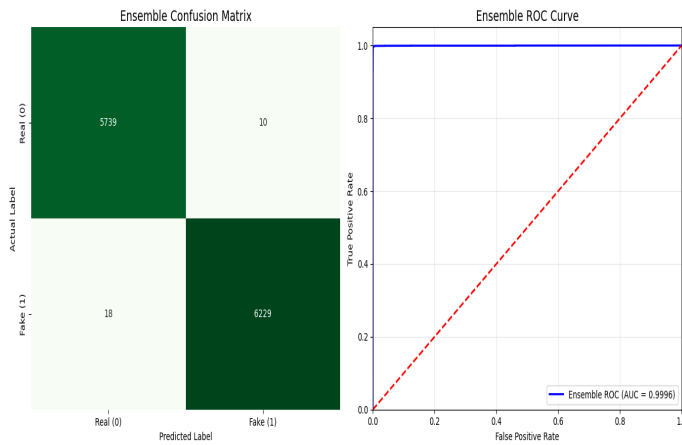


Fig. 5: Dataset 1 confusion matrices for the Ensemble

"Fake" class (0.71), its precision was very poor (0.49). This indicates that the model wasn't necessarily learning; rather, it was biased toward predicting everything as "Fake".

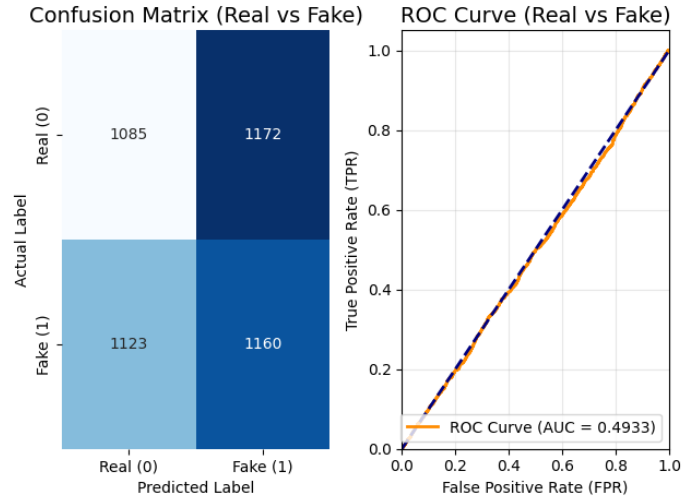


Fig. 8: Dataset 2 confusion matrices for the XGBoost

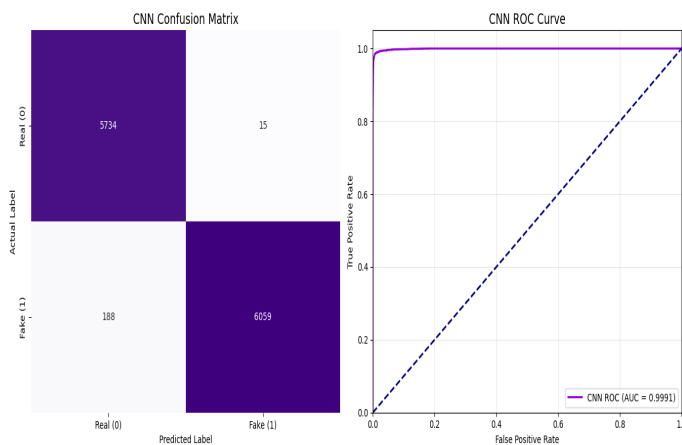


Fig. 6: Dataset 1 confusion matrices for the CNN

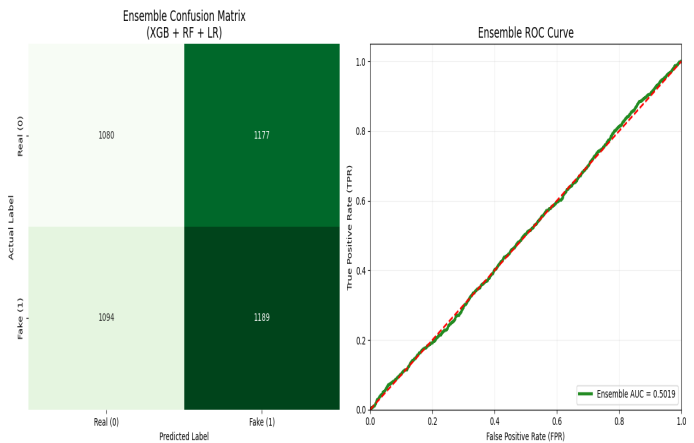


Fig. 9: Dataset 2 confusion matrices for the Ensemble

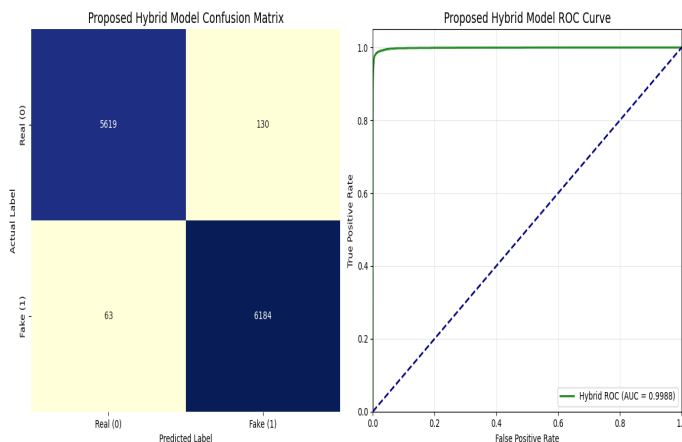


Fig. 7: Dataset 1 confusion matrices for the Proposed Hybrid(CNN-XGB)

In Dataset 2, the performance plateaued across the board, suggesting that the models couldn't find distinct predictive features. While the CNN model showed a higher recall for

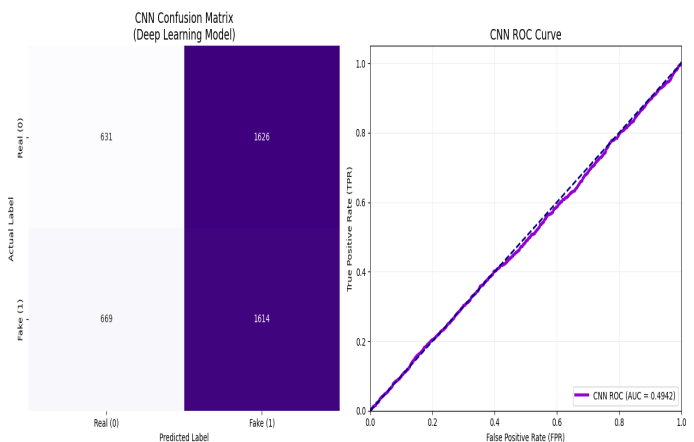


Fig. 10: Dataset 2 confusion matrices for the CNN

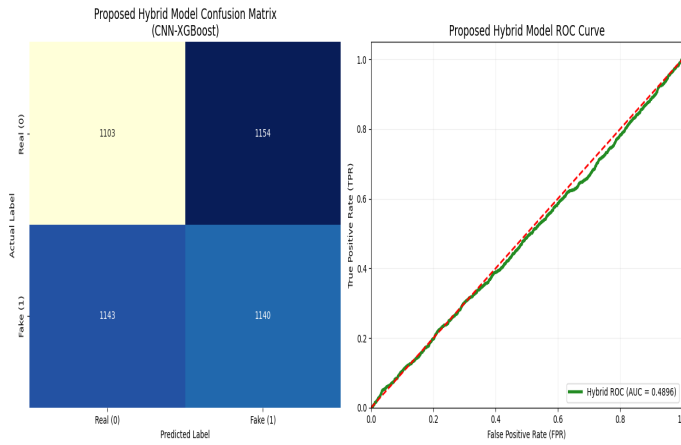


Fig. 11: Dataset 2 confusion matrices for the Proposed Hybrid(CNN-XGB)

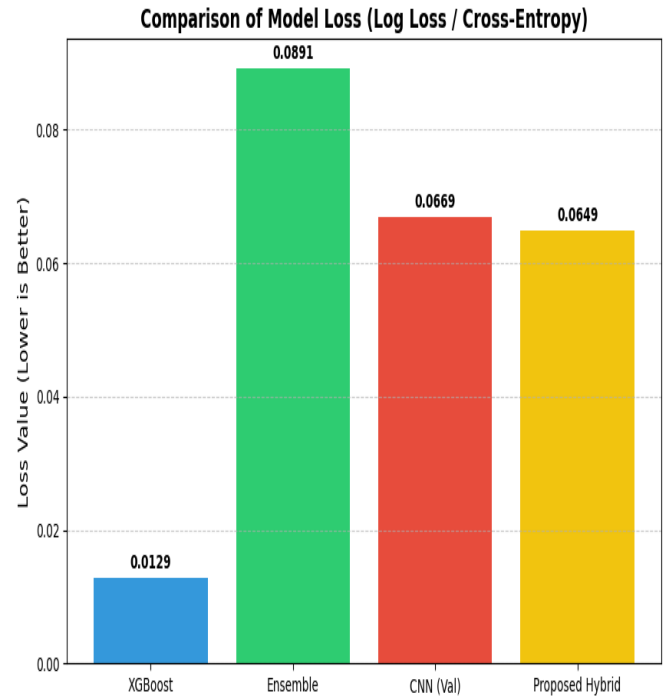
D. Loss Function and Model Confidence

Beyond just looking at right and wrong answers (accuracy), we dug into the Log Loss to see how "confident" the models actually were in their predictions. This gives us a peek under the hood: is the model making a solid choice, or is it just guessing?

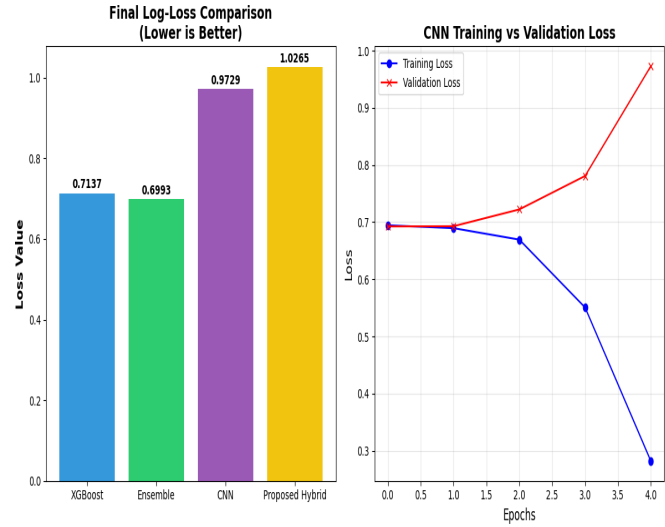
For Dataset 1, the story was all about high confidence. XGBoost didn't just perform well; it was incredibly sure of its decisions, clocking in a tiny Log Loss of just 0.0129. We also noticed that our Proposed Hybrid Model achieved a lower loss (0.0649) compared to the standard CNN (0.0669). This proves that stacking the XGBoost classifier on top of the CNN helped stabilize the decision-making process, making the model less "jittery" than the deep learning model alone.

However, things fell apart when we switched to Dataset 2. The loss values shot up past 0.69, which is basically the mathematical equivalent of flipping a coin.

- The Ensemble model hovered right around this guessing point (0.6993), meaning it was effectively stumped by the data.
- The Hybrid model struggled the most here, with the loss spiking to 1.0265. This high number suggests the model wasn't just wrong—it was confidently wrong. This happens when the underlying CNN over-fits the training data and feeds misleading, "noisy" features into the classifier.



(a) Dataset 1 Loss (High Confidence)



(b) Dataset 2 Loss (Low Confidence/Overfitting)

Fig. 12: Side-by-side comparison of Log Loss metrics. Figure (a) shows extremely low loss (high confidence) for Dataset 1, while Figure (b) illustrates significantly higher loss and overfitting for Dataset 2.

E. Error Analysis and Overfitting

The most telling data comes from the loss metrics, which explain why the models failed on the second dataset.

- **Loss Explosion:** For XGBoost, the Log Loss jumped from a confident 0.0129 in Dataset 1 to 0.7137 in Dataset 2, showing that the model became highly uncertain of its predictions.

- **The Overfitting Trap:** The training graphs for the CNN on Dataset 2 clearly show the model diverging around Epoch 3. Even though the training loss kept dropping to 0.2931, the validation loss skyrocketed to 0.9729. This is a classic sign that the model was memorizing noise instead of learning generalizable patterns
- **Hybrid Model Issues:** Consequently, the Hybrid model suffered the highest Log Loss (1.0265) because it relied on feature extraction from an overfitted CNN, feeding poor data into the XGBoost classifier.

F. Conclusion on Robustness

Ultimately, while our proposed models proved highly effective on Dataset 1, they struggled to replicate that success on Dataset 2. The inconsistent results suggest that the features learned in the second environment were not robust. Among all approaches, the Ensemble model remained the most stable, though even it could not overcome the underlying data challenges present in the second dataset.

VI. DISCUSSION

A. Interpretation of Findings

Our experiments painted two very different pictures, highlighting just how much machine learning models rely on the quality of the data they are fed.

- **The Performance Gap:** On Dataset 1, the results were straightforward. The Ensemble and XGBoost models achieved nearly perfect accuracy (99.77%). This suggests that the dataset had clear, distinct linguistic patterns—like specific keywords or phrasing—that made it easy for the models to draw a line between "Real" and "Fake" news.
- **The Overfitting Problem:** Dataset 2, however, was a completely different story. Here, every single model hit a wall, hovering around 50% accuracy. The most telling sign was in the CNN's learning curve (Figure 12); while the model seemed to be learning (training loss dropped), its performance on new data got worse (validation loss spiked). This is a textbook case of overfitting—the model was essentially "memorizing" the noise in the training data rather than understanding the actual content.
- **Hybrid Model Behavior:** Interestingly, our Proposed Hybrid Model (CNN-XGBoost) showed its strength in Dataset 1 by fixing the class imbalance. It was much sharper at catching fake news, missing only 63 cases compared to the 188 missed by the standard CNN. But in Dataset 2, this same architecture backfired, showing the highest Log Loss (1.0265). This happened because the CNN feature extractor was feeding poor, overfitted data into the XGBoost classifier, which just made the model more confident in its wrong answers.

B. Limitations

- **Data Sensitivity:** The study clearly shows that our framework lacks domain adaptation. It works perfectly when the text follows a specific structure (Dataset 1) but fails

when the writing style becomes noisier or more subtle (Dataset 2).

- **Static Features:** By relying on TF-IDF and basic CNN embeddings, the models missed out on the deeper "context" that modern methods (like Transformers) might catch. They looked for words, not meaning.
- **Deep Learning Instability:** Without heavy regularization, the deep learning layers were too prone to overfitting on the challenging second dataset.

C. Possible Improvements

To fix the instability we saw in Dataset 2, future iterations need to focus on:

- **Smarter Embeddings:** Swapping out TF-IDF for pre-trained models like BERT or RoBERTa, which understand context better than simple keyword matching.
- **Fighting Overfitting:** We need to be more aggressive with regularization techniques—like higher dropout rates or Early Stopping—to stop the CNN from memorizing the training data.
- **Data Augmentation:** We could artificially expand the noisy dataset using synonym replacement or back-translation to force the model to learn actual semantic patterns rather than noise.

VII. CONCLUSIONS

In this study, we explored how well different machine learning approaches—from classical trees to deep neural networks—could detect fake news. We also introduced a Hybrid CNN-XGBoost framework specifically designed to combine the feature-learning power of CNNs with the precision of XGBoost.

A. Summary of Contributions:

- **Hybrid Success in Controlled Environments:** We demonstrated that when the data is structured, our Hybrid approach effectively improves recall. It creates a more balanced system that is less likely to let fake news slip through the cracks compared to a standalone CNN.
- **The Stability of Ensembles:** Overall, the Ensemble model proved to be the safest bet, offering the most consistent performance (99.77% on Dataset 1) across the board.
- **A Cautionary Tale on Generalization:** Most importantly, the drastic failure on Dataset 2 serves as a crucial finding. It proves that high accuracy on one dataset doesn't guarantee a robust model. Our analysis of the Log Loss showed that without proper safeguards, deep learning models can easily fall into the trap of overfitting when faced with noisy, unstructured data.

B. Future Directions:

Moving forward, our goal is to bridge this generalization gap. We plan to integrate Transfer Learning (using Fine-tuned BERT) to handle noisy data better. We also want to implement Explainable AI (XAI) tools to visualize exactly

why the models failed on Dataset 2, helping us build a system that isn't just accurate, but also trustworthy and transparent.

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